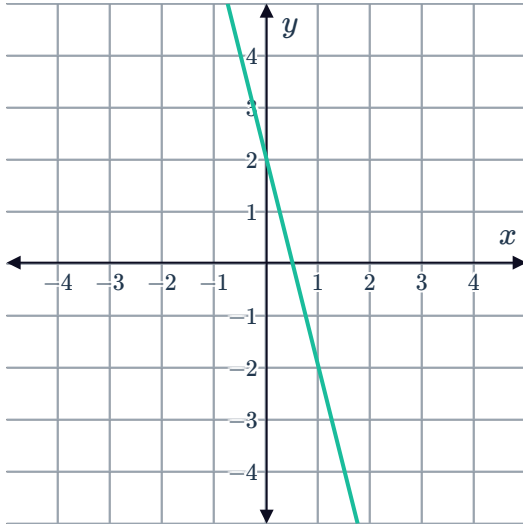


Transformations of $y=x$

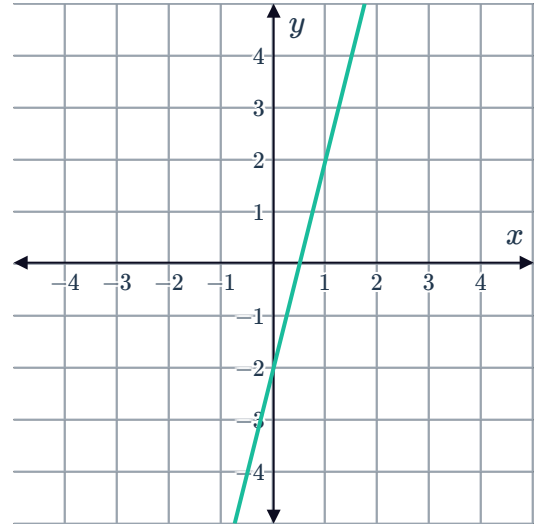


1 Determine whether the following graphs are decreasing linear function.

a

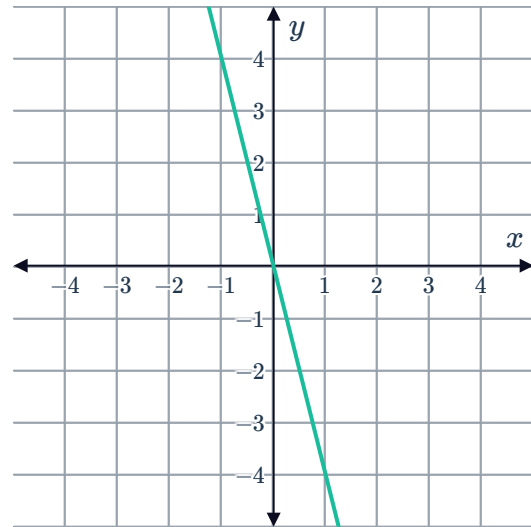


b



2 Consider the graph of $y = -4x$.

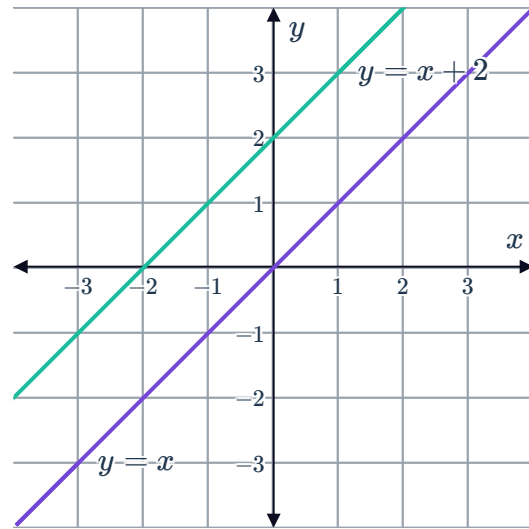
- a As x becomes larger in the positive direction (i.e. x approaches ∞), what happens to the corresponding y -values?
- b As x becomes larger in the negative direction (i.e. x approaches $-\infty$), what happens to the corresponding y -values?



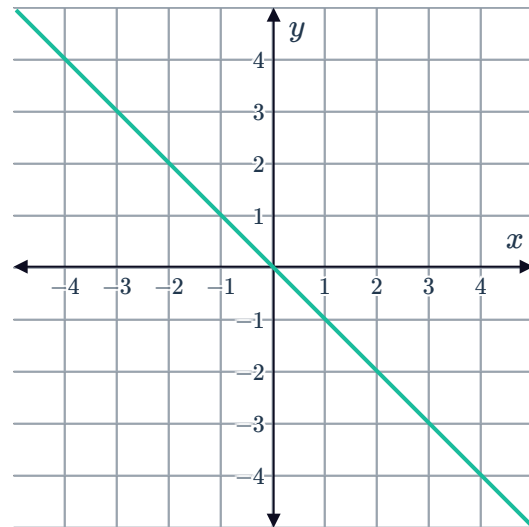
- 3 The lines $y = x + 2$ and $y = x$ have been graphed on the same coordinate plane. Determine whether the line $y = x + 2$ is:



- a Increasing or decreasing.
- b Steeper, flatter or equally steep as $y = x$.

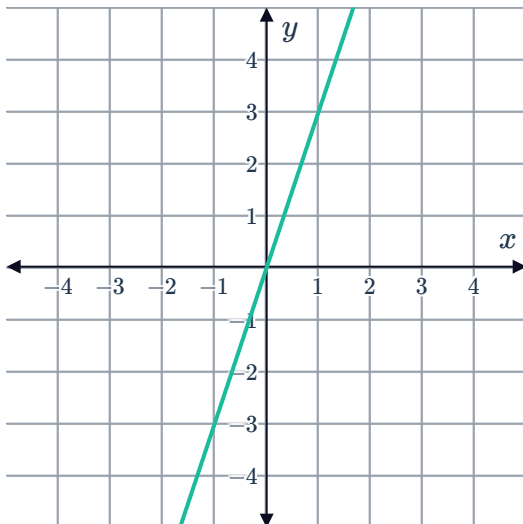


- 4 Consider the given graph of $y = -x$.

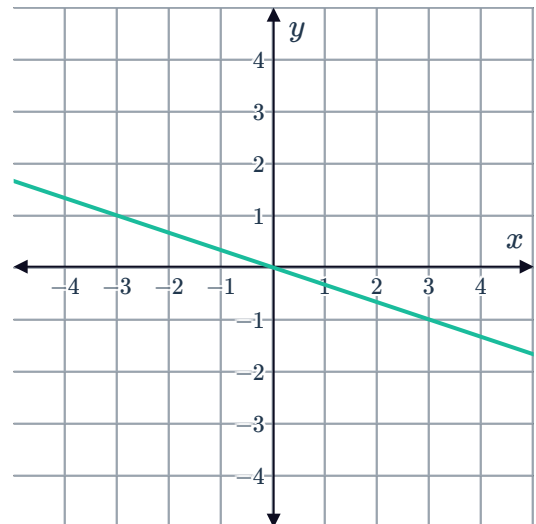


Determine whether the following shows the graph of a straight line whose y value is decreasing at a faster rate than $y = -x$.

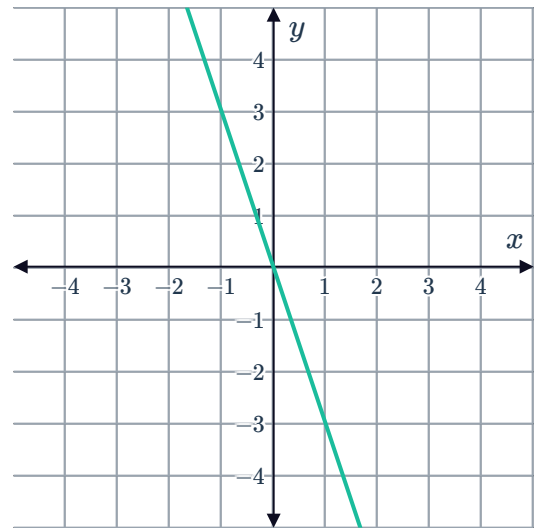
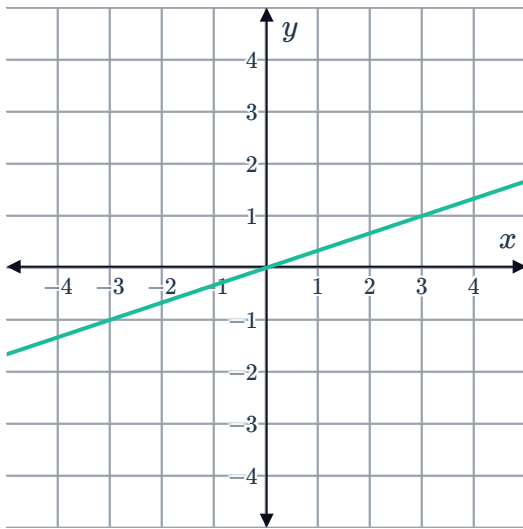
a



b



c

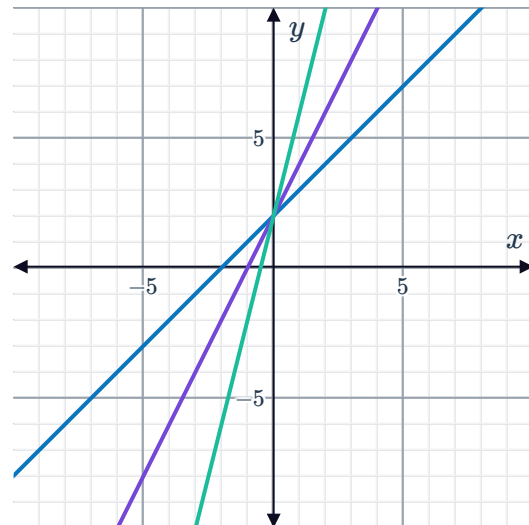


5 A straight line graph has a positive y -intercept and a positive x -intercept. Determine whether the following must also be true of the straight line graph.

- a The straight line graph is flatter than $y = x$.
- b The straight line graph is increasing.
- c The straight line graph is steeper than $y = x$.
- d The straight line graph is decreasing.

6 The equations: $y = x + 2$, $y = 2x + 2$ and $y = 4x + 2$ have been graphed on the same number plane:

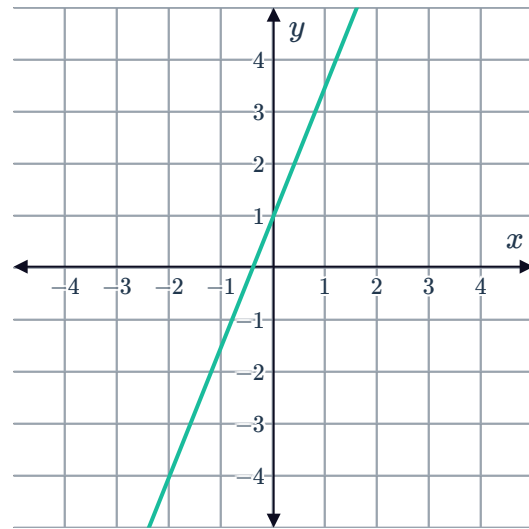
- a What do all of the equations have in common?
- b What do all of the graphs have in common?
- c What can you conclude from the answers above?



7 Look at the graph of the line.



- a At the y -intercept, $x = 0$.
What is the y -value at this point?
- b The equation of the line is $y = \frac{5x}{2} + 1$.
For a straight line equation of the form $y = mx + b$, which variable represents the y -intercept?

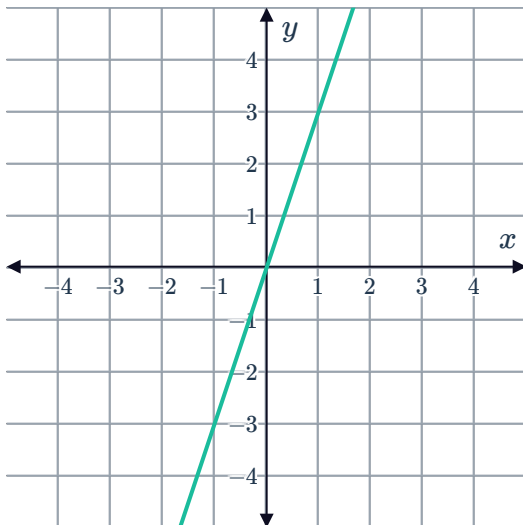


8 Consider the following graphs:

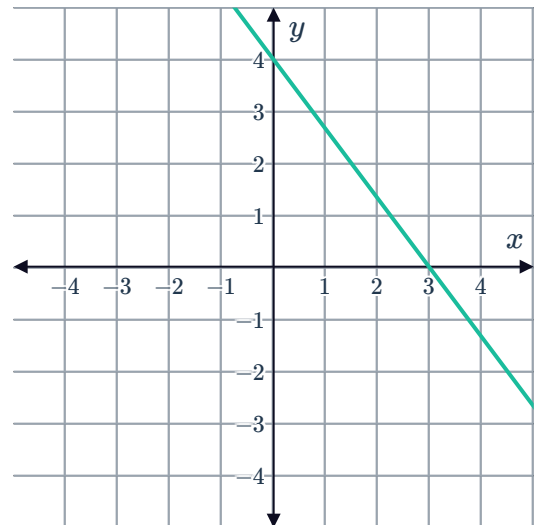
i State the x -value of the x -intercept.

ii State the y -value of the y -intercept.

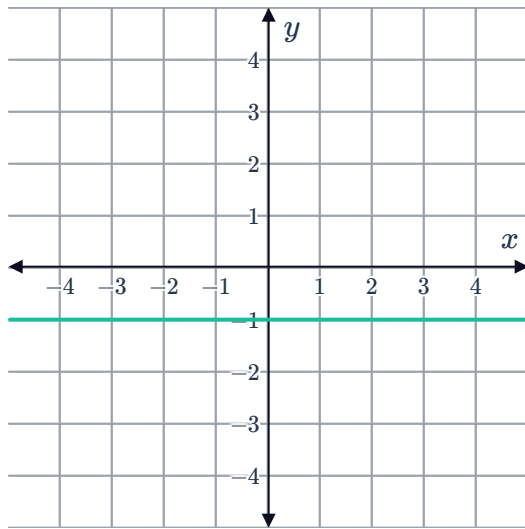
a



b



c



9 Determine whether the following equations represent lines that will intersect the y -axis at 2.

a $y = 2x - 5$

b $y = 2x$

c $y = 2 - x$

d $y = 5x + 2$

e $y = x - 2$

f $y = x + 2$

10 A linear graph has a negative y -intercept and a positive slope.
Is the x -intercept negative or positive?

11 Determine whether the following lines pass through the origin.

a $y = 6x$

b $y = \frac{x}{6}$

c $y = 6x - 6$

d $y = 8x + 8$

e $y = 8x$

12 Find the y -value of the y -intercept for the following.

a $y = -7x + 5$

b $y = 5x$

c $y = 10 - 5x$

d $y = -x + 1$

13 Determine whether the following lines have a y -intercept.

a $y = x$

b $x = 3$

c $y = -3$

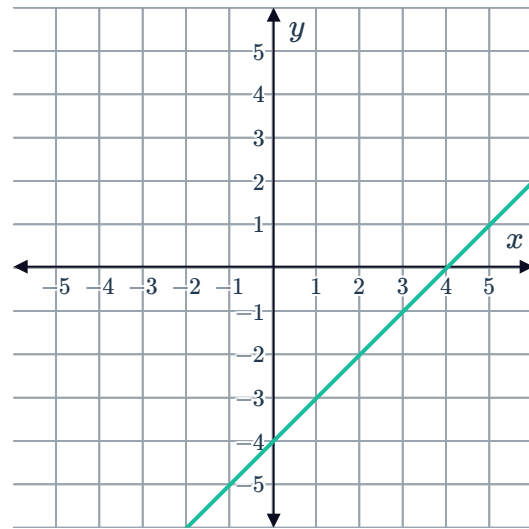
d $3x - y = 0$

e $y = 4x + 5$

14 Consider the graph of $y = x - 4$.

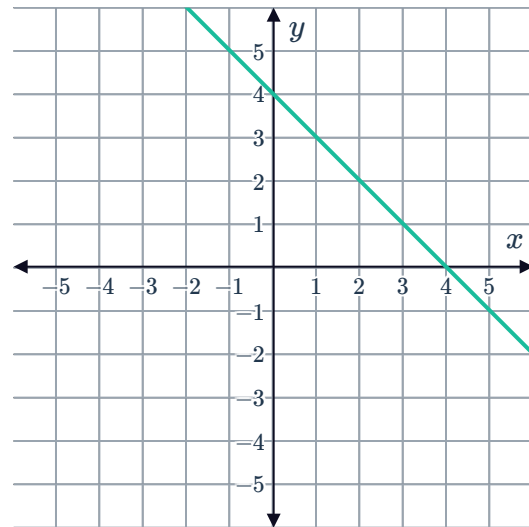


- a State the coordinates of the x -intercept.
- b State the coordinates of the y -intercept.
- c If the graph is translated 6 units down, find the coordinates of the new y -intercept.

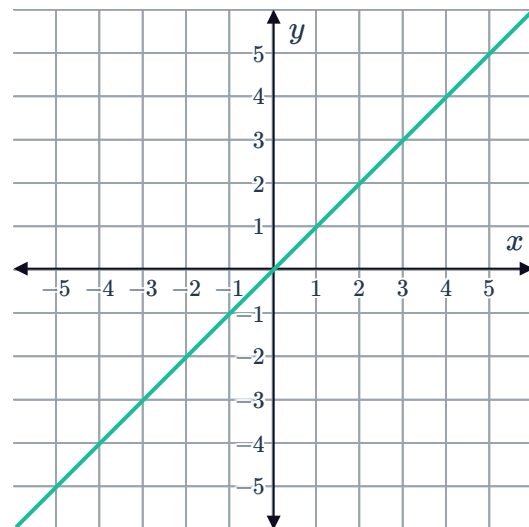


15 Consider the graph of $y = 4 - x$.

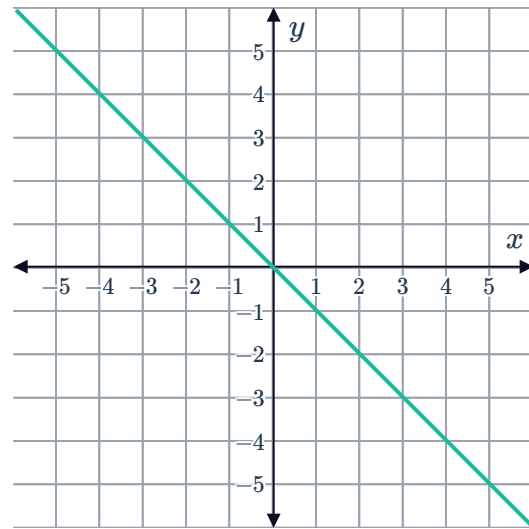
- a State the coordinates of the x -intercept.
- b State the coordinates of the y -intercept.
- c If the graph is translated 10 units up, find the coordinates of the new y -intercept.



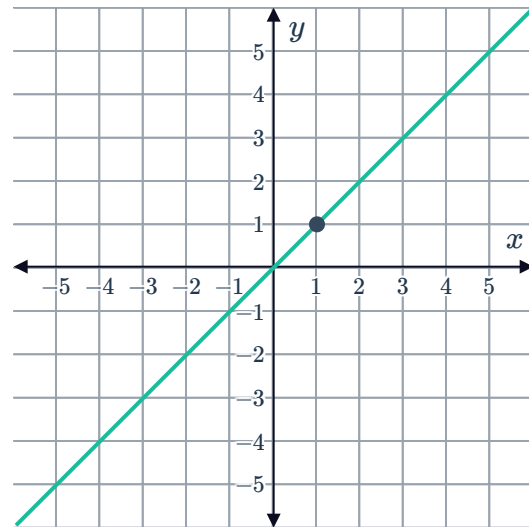
16 Consider the graph of $y = x$. Draw the graph of the resulting line translated 2 units down.



- 17 Consider the graph of $y = -x$. Draw the graph of the resulting line translated 4 units down.

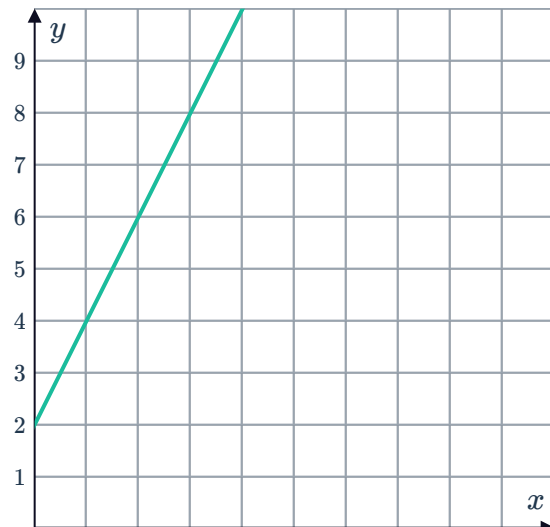


- 18 Consider the graph of $y = x$, with the point $(1, 1)$ that lies on the graph. The steepness of this graph is to be quadrupled. Draw the graph of the resulting line.



- 19 Part of a continuous straight line graph has been graphed. Determine whether each of the following statements is true about the entire line graph.

- a The points on the graph have both positive and negative y values.
- b The graph has a negative x -intercept.
- c The graph passes through the origin.
- d The points on the graph have only positive y values.
- e It has a negative slope.

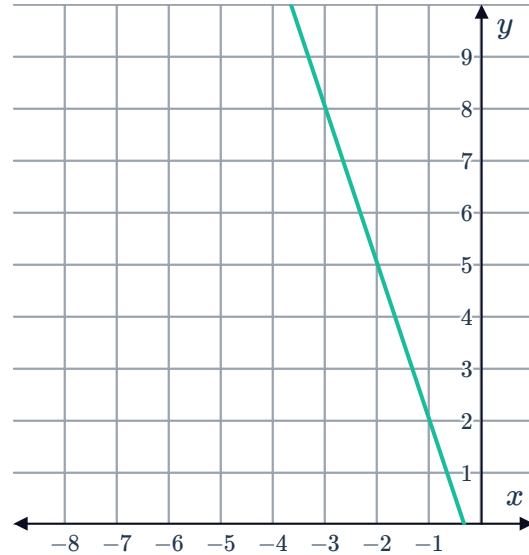




20 Part of a straight line graph has been graphed.

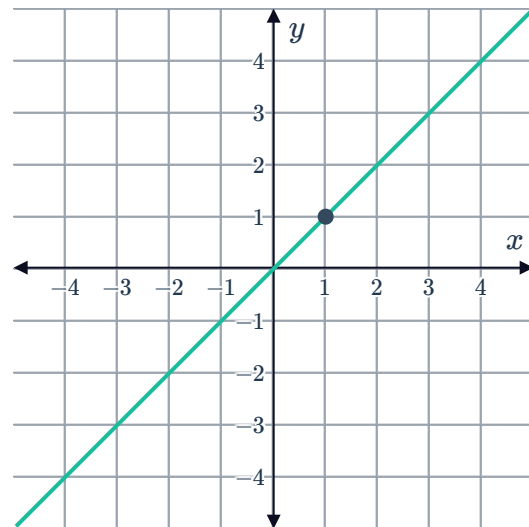
Determine whether each of the following statements is true of the entire line graph.

- a** The points on the graph have both positive and negative y values.
- b** The points on the graph have only negative y values.
- c** It has a negative slope.
- d** The graph is increasing.
- e** The graph has a positive y -intercept.
- f** The graph passes through the origin.



21 Consider the graph of $y = x$.

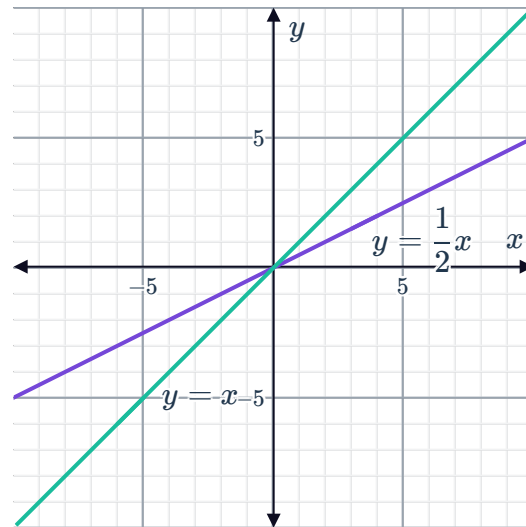
Where will the point $(1, 1)$ move to if the line is made 3 times as steep?



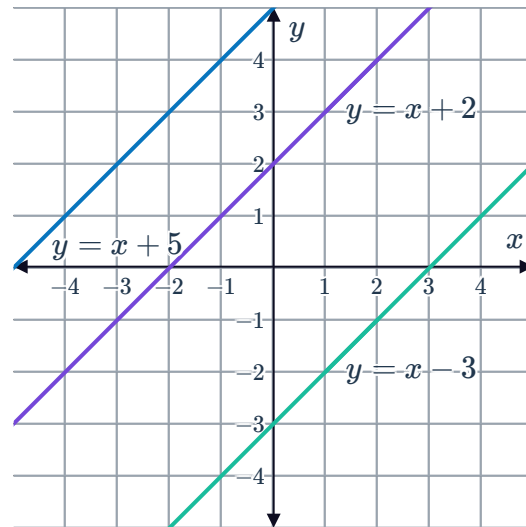
22 The lines $y = \frac{1}{2}x$ and $y = x$ have been graphed in the same set of axes.



- a Is the line $y = \frac{1}{2}x$ increasing or decreasing?
- b Is it steeper, flatter or equally steep as $y = x$?
- c Determine whether the following could be done to transform the graph of $y = x$ into $y = \frac{1}{2}x$.
 - i Reflecting $y = x$ about the x -axis.
 - ii Reflecting $y = x$ about the y -axis.
 - iii Making $y = x$ steeper.
 - iv Making $y = x$ flatter.



23 Consider the graph of the lines $y = x + 5$, $y = x + 2$ and $y = x - 3$.



- a Determine whether the following statements are true.
 - i They are all the same steepness.
 - ii All lines that are parallel to $y = x$ have positive y -intercepts.
 - iii All lines that are parallel to $y = x$ have positive x -intercepts.
 - iv They can all be made to overlap $y = x$ through a vertical translation.
 - v They are all parallel to one another.

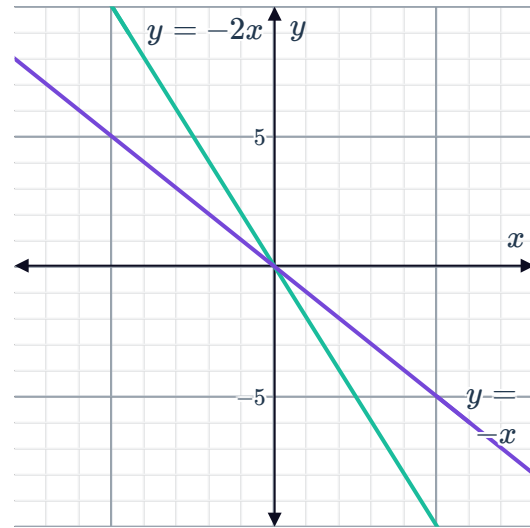
b Determine whether the following are true or false about the x - and y -intercepts of lines that are parallel to $y = x$.



- i** Each line has x and y intercepts that are equal.
- ii** The x and y intercepts of these lines are the same distance from the origin.
- iii** Each line has x and y intercepts that are equal in magnitude, but opposite in sign.

c If a straight line is parallel to $y = x$ and has a y -intercept of -8 , find its x -intercept.

24 The lines $y = -2x$ and $y = -x$ have been graphed on the same set of axes.



- a** Is the line $y = -2x$ increasing or decreasing?
- b** Is it steeper, flatter or equally steep as $y = -x$?
- c** Here are some transformations.
 1. Reflection about the x -axis.
 2. Steepening of straight line.
 3. Reflection about the y -axis.
 4. Translation 2 units down.
 5. Flattening of straight line.
 6. 90° rotation about the origin.

Determine whether the following pairs of transformations turn the graph of $y = x$ into the graph of $y = -2x$.

- i** Transformations 1 and 2
- ii** Transformations 2 and 3
- iii** Transformations 1 and 5
- iv** Transformations 2 and 6
- v** Transformations 4 and 5

25 Which of the following statements is true for each of the following set of points.



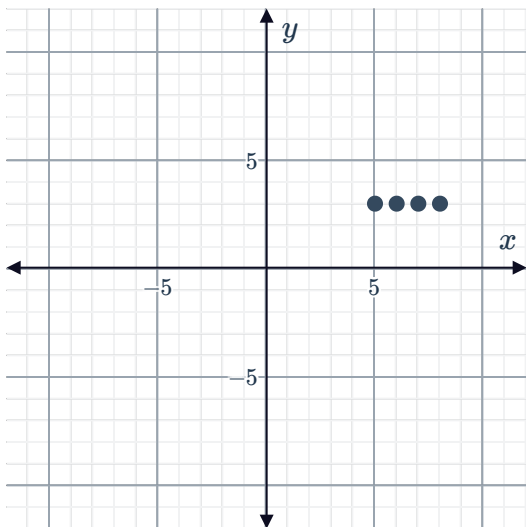
Statement 1: The set of points lie on a vertical line.

Statement 2: The set of points lie on a horizontal line.

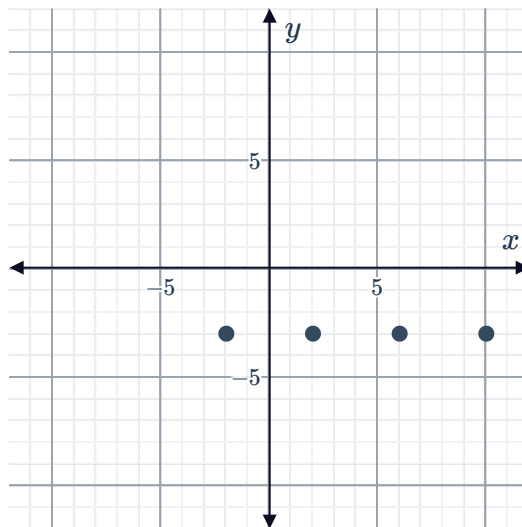
Statement 3: The set of points lie on an increasing line.

Statement 4: The set of points lie on a decreasing line.

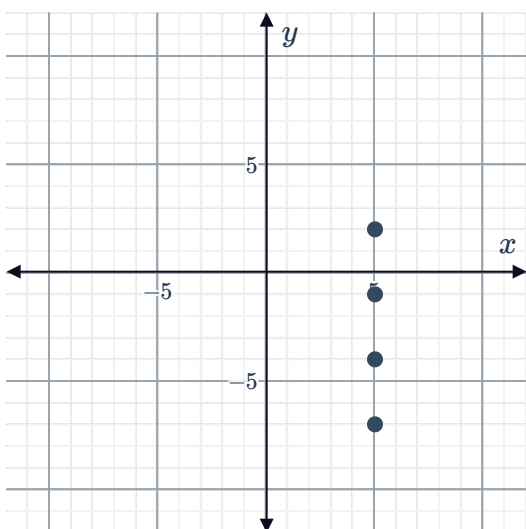
a



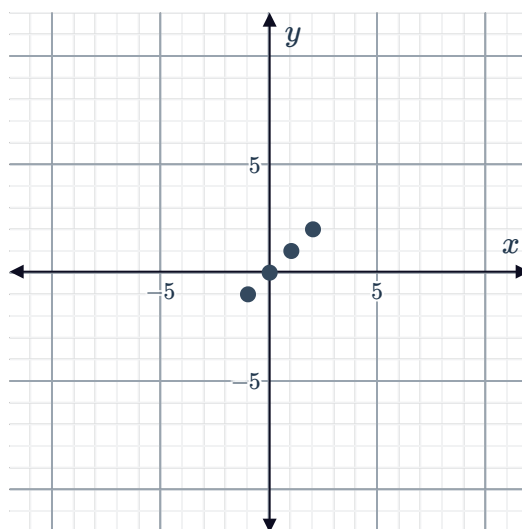
b



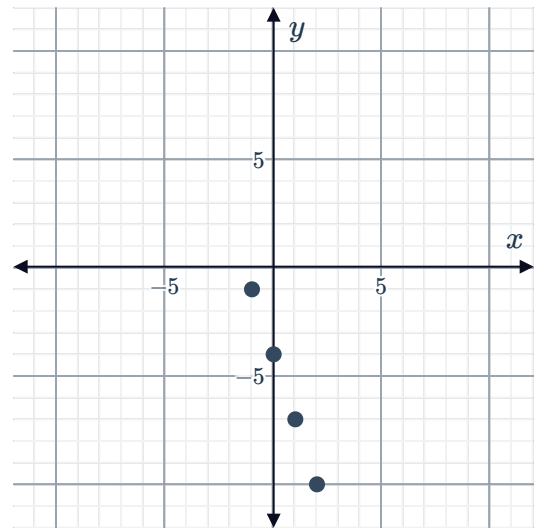
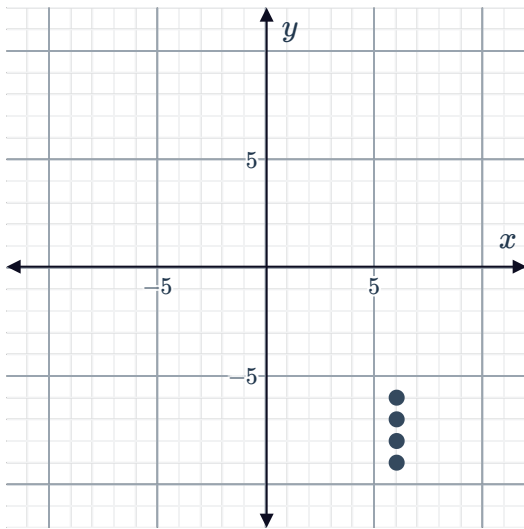
c



d



e



26 Determine whether the graph of the following are horizontal or vertical line.

a $y = 6$

b $x = -4$

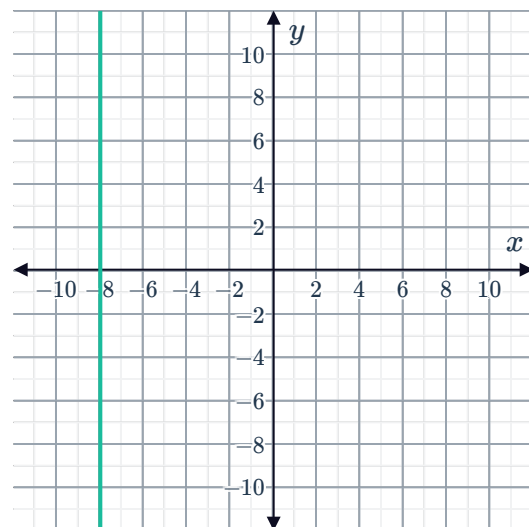
27 Plot the graph of the following lines:

a $y = -7$

b $x = -8$

28 Consider the graphed line.

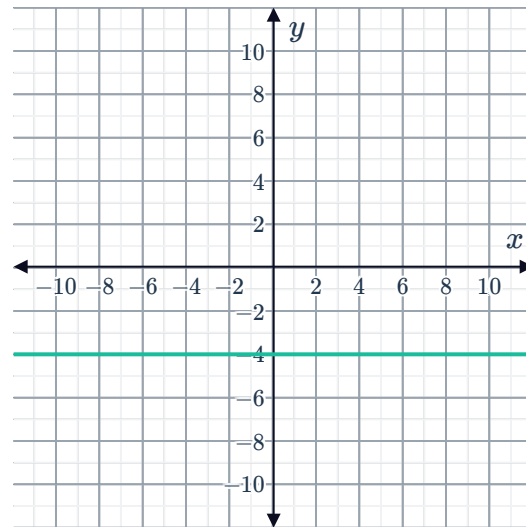
- a** State the x -value of the x -intercept of the line.
- b** How many y -intercepts does the line have?



29 Consider the graphed line.

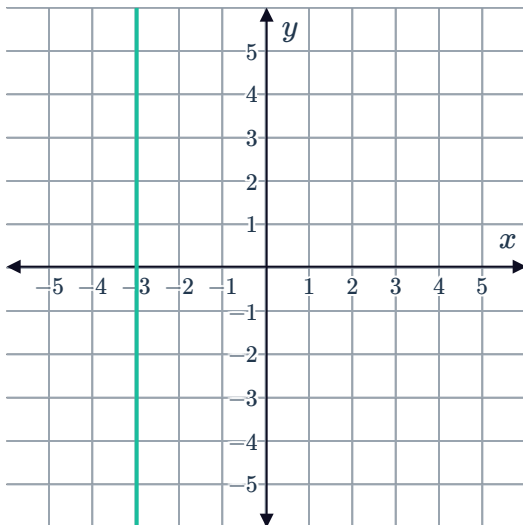


- Determine the y -value of the y -intercept of the line.
- How many x -intercepts does the line have?

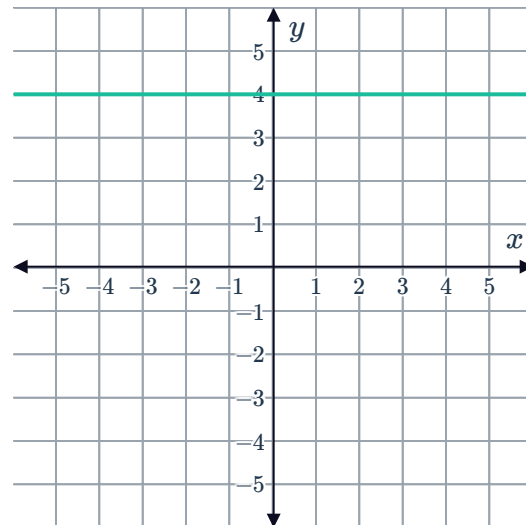


30 Determine the equation of the following graphed lines.

a



b



31 Consider the table of values shown.
Find the equation of the line which passes through these points.

x	-4	-2	0	2
y	-4.5	-4.5	-4.5	-4.5

32 The table shows some points on the line $y = 0$.
The line $y = 0$ represents which axis?

x	-4.5	-0.5	3.5	8.5
y	0	0	0	0

- 33** The table shows some points on the line $x = 0$.



x	0	0	0	0
y	-5.5	-2.5	3.5	6.5

The line $x = 0$ represents which axis?

- 34** Find the equation for the following:

- a** All horizontal lines that are 10 units away from the line $y = 3$
- b** All vertical lines that are 7 units away from the line $x = -2$

- 35** Three of the lines that enclose a square on a number plane have equations $x = 0$, $x = 2$ and $y = -2$.
Find all the possible equations of the fourth line.

- 36** The cost of a taxi ride C , in dollars, is given by $C = 2.5t + 2$, where t is duration of trip in minutes.
- a** For every extra minute the trip takes, how much more will the trip cost?
 - b** State what the constant value 2 could represent.

Additional questions

- 37** Describe the transformation of $f(x)$ that is of the form:

- a** $g(x) = f(x) + k$
- b** $g(x) = kf(x)$
- c** $g(x) = -f(x)$

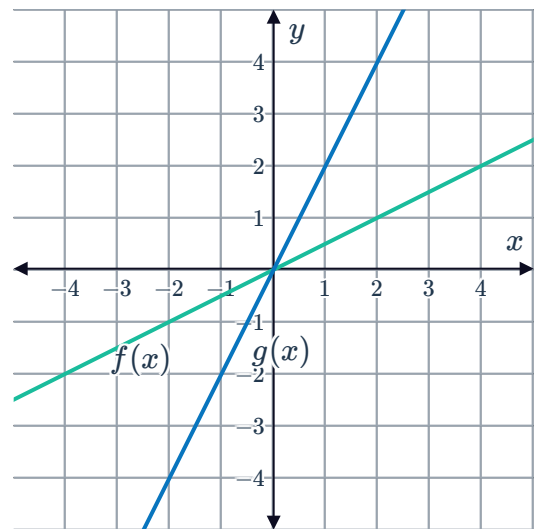
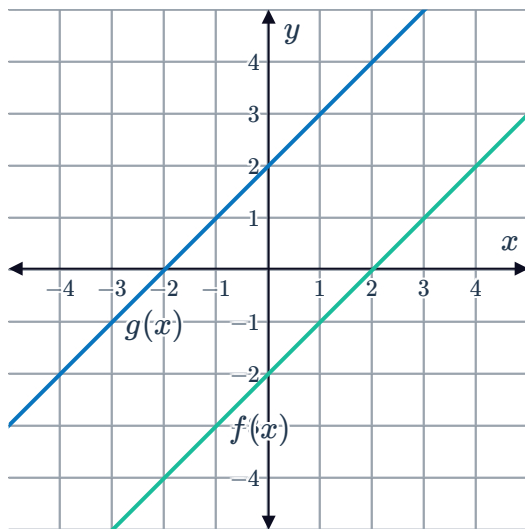
- 38** For the what values of k will the transformation from $f(x)$ to $g(x) = kf(x)$ involve a reflection across the x -axis?

- 39** For each of the following, describe the type of transformation from $f(x)$ to $g(x)$.

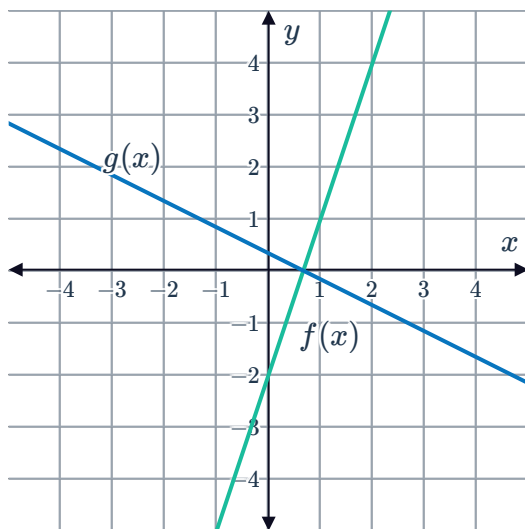
- a** $g(x) = f(x) + 4$
- b** $g(x) = 6f(x)$
- c** $g(x) = \frac{1}{3}f(x)$
- d** $g(x) = \frac{5}{12}f(x)$
- e** $g(x) = -f(x)$
- f** $g(x) = f(x) - 2$

- 40** For each of the graphs, describe the type of transformation from $f(x)$ to $g(x)$.

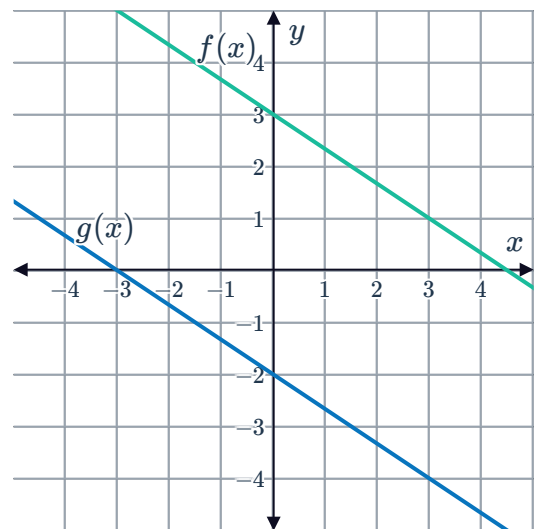
a



c

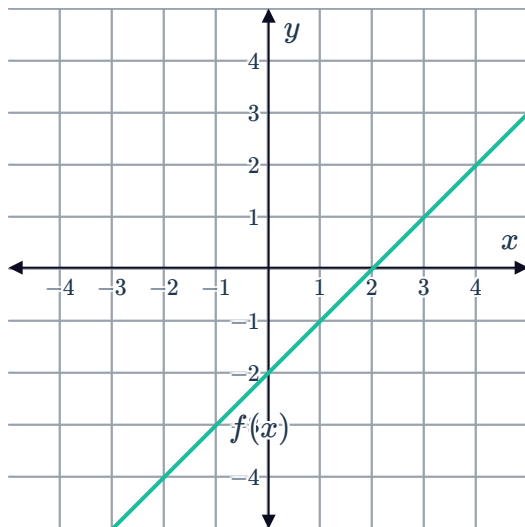


d

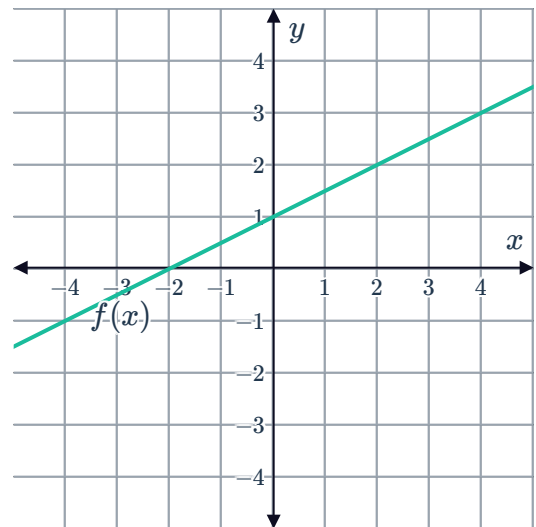


41 For each graph of $f(x)$, draw the graph of $g(x)$ using the given transformation.

a $g(x) = f(x) + 5$



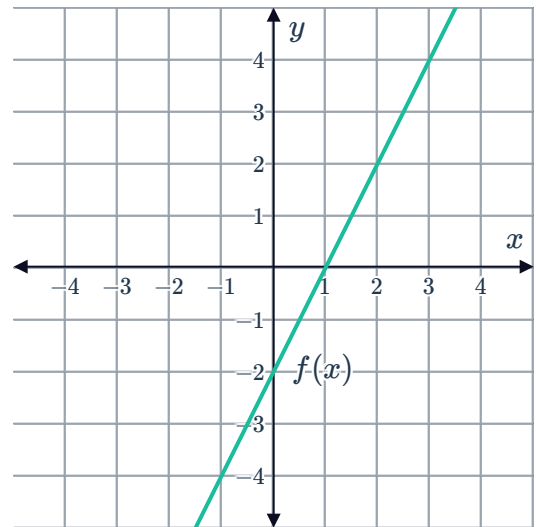
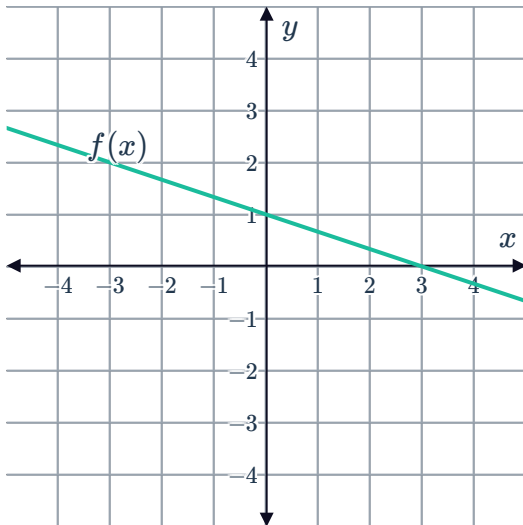
b $g(x) = 2f(x)$



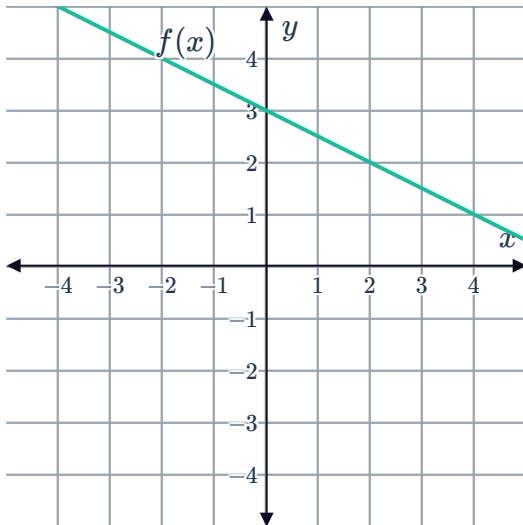
c $g(x) = -f(x)$



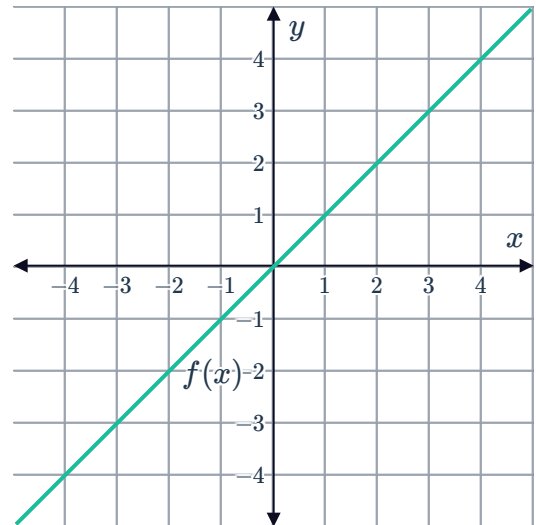
$g(x) = -\frac{1}{4}f(x)$



e $g(x) = f(x) - 2$



f $g(x) = f(x) + 2$



42 For each transformation, describe how the graph of $g(x)$ is related to the graph of $f(x)$.

a $g(x) = f(x) - 8$

b $g(x) = f(x) + \frac{1}{2}$

c $g(x) = -3f(x)$

d $g(x) = \frac{1}{4}f(x)$

43 Oreste draws the graph of $g(x) = -f(x)$, as shown.



- a Describe the error Oreste made.
- b Describe the type of graphs that $f(x)$ could be for Oreste's error to still give him the correct answer.

